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A Detailed Analysis of the Critical Role of artificial intelligence in Enabling High-Performance Cloud Computing Systems

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***Abstract*- Cloud computing reflects effective delivery of computing services such as software, storage purposes, analytics, intelligence and many others. Besides that, it also offers faster innovation and at the same time flexible resources. In an organisation, a cloud computing system provides tight security of data with a low maintenance cost. A huge collaboration has been seen between various industries through cloud systems. Those all processes have been boosted up by the implementation of AI. Incorporation of AI by those industries hugely influences the efficiency rate of cloud computing techniques and due to this reason, the combined effect of those two technologies brings up huge profitability to those organisations. Implementation of machine learning models, as well as cognitive computing, increases the performance of cloud systems. In addition to this, a lot of benefits have been seen when organisations incorporate AI in cloud computing techniques.**

This paper contains a detailed introduction part along with a well informative literature review. A research methodology also has been provided where three research questions are mentioned.

The data analysis part comprises a multi regression analysis where the independent variables are years and percentage of usage of AI in various organisations. The dependent variable is profitability. A critical discussion part also has been presented and a conclusion has also been provided with a few recommendations.

Keywords- Artificial intelligence, cloud computing, multi regression analysis, machine learning techniques, data security, cost-effectiveness, cognitive computing technique

I. INTRODUCTION

Artificial Intelligence defines some special and unique ability of any computer that is controlled by another computer to effectively perform several important tasks. Artificial intelligence highly requires human intelligence such as visual perception, recognition of speech, making of decisions and many others [1]. In recent days this system utilises in various important sectors such as monitoring of social networking sites, investigation of financial processes, healthcare management and many others. This intelligence system also plays a significant role in ensuring the high performance of cloud computing systems. On the other hand, cloud computing reflects the delivery of various kinds of services by using internet systems. This technology has a wide range of resources such as various databases, software, servers and many others. Through this cloud computing technology, those resources can be easily utilised and a detailed view of software can also be gained easily through this process. Data backup process, development of software, analysis of big data is some of the important things that can easily be handled through this technology. This research paper is going to be explored a transparent evaluation of the role of artificial intelligence that helps develop high cloud computing systems through using multi regression analysis. The multi regression analysis technique is a specific statistical analysis tool that can be utilised for evaluating the relationship between several dependent and independent variables [2]. In this paper, artificial intelligence and cloud computing systems can be considered as two important variables and those two tools can be effectively analysed through this technique. Artificial intelligence (AI) supported cloud systems utilises various data and based on that information, it makes predictions. Various important problems can be easily solved before noticing by the users. Some importance of AI-based technology that benefits cloud computing is given in the following.

II. LITERATURE REVIEW

According to Holzinger artificial intelligence is probably the oldest technology in the domain of computer science. It was forecasted that almost ninety per cent of the organisation would employ AI in their business initiative. It is also forecasted at the same time that those organisations would also incorporate AI in their cloud system. It is predicted that around seventy per cent of the organisations would import their data through effective employment of tools. The data

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shows that around sixty-five per cent would start to formulate applications in this particular scenario. Even in the domain of 2021, individuals believe that AI is an apparatus for the future. Almost all the individuals with a smartphone in their hand tend to employ cloud computing extensively. The effectiveness of AI tends to go unnoticed as it plays a major role in the statistical prediction in a particular scenario. Cloud computing features such as dynamic, regulated sharing of data pools of computational services have enabled the technology development and frameworks to meet the needs of growing applications such as smart cities, scientific research, agriculture, healthcare, as well as traffic management. However, it is important to comprehend the correlation between cloud and AI can vehemently transform the pattern on which an organisation operates in their business scenario. It can maximise the efficiency into ten folds. Different types of technological advancement can occur due to the efficient marriage of AI and clouds. The core technologies for employing cloud computing are distributed computing, internet tools, system administrative technologies and hardware technologies. It can effectively revolutionise the functioning of the business. It enables the organisation to tap into machine learning assets assimilated with the smart cloud. It can be compartmentalised into two distinct categories. They are Cloud AI initiatives and Machine learning programmes. This robust collaboration effectively provides the organisation with a more assertive, well-organised and unlimited database to store the big data and employ them systematically. Additionally, the assimilation between AI and cloud computing influences the initiatives as it systematically helps to manage automatic transaction recognition, anomaly optimisation, allocation identification, predictive emulation, aggressive development and many others. Multiple regression examination and analysis plays a key role to predict the influence of employing cloud computing for different organisations. It is basically a technique that assists to analyse the correlation between a single determinant and multiple determinants. The core aim of the multiple regression is to analyse the value of a single variant. It can effectively predict the influence of cloud computing in an organisation.

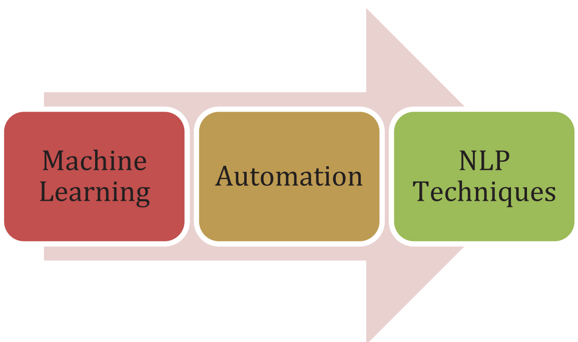

Figure 1: Techniques of AI
(Source: [4])

According to Malik Cloud computing is regarded as the technology that will shape the future efficiently. He also provided a broad perspective on cloud computing while shedding a light on the technologies associated with it. It also illuminated virtualisation in the journal. Cloud computing is basically a terminology that has become vehemently popular in the last decade. It generally enables individuals to tap into all the attributes of the said system while cancelling the bulk into the core system. Numerous people tend to employ this cloud computing technology without even comprehending it. This technology enables individuals to store their personal information in a "cloud-hosted server" that efficiently stockpile the data for future usage. It plays a vital role in different kinds of initiatives as it helps organisations to manage a large amount of data. It has multiple benefits. It vehemently minimises the expenditure spent on information technology. It includes minimisation of the cost of system manipulation [5]. It also tends to maximise the scalability of the initiatives. It effectively frees up time consumption. It assists to run the business effectively. It also assists in business continuity. It includes natural disasters and other unwanted scenarios. It tends to maximise the collaboration efficacy as it provides the opportunity to communicate efficiently and put forward the box idea. It provides flexibility in the working scenarios. It provides the opportunity for gaining automatic updates. Cloud computing has become a significant resource across many parts of society, including academics, government, and business, as a result of its fast expansion. Cloud computing features include dynamic, regulated access to pooled pools of computer resources. Sensors and actuators, as well as other IoT elements, interface with the gateway, which is all networked through 5G networks. The users can interact with the model using an app application on the access point, which uses Bitcoin (or an equivalent cryptocurrency) software to estimate costs. Cloud computing is a new paradigm that allows for on-demand, regulated access to computational capacity, allowing for technological advancement and geographically dispersed applications.

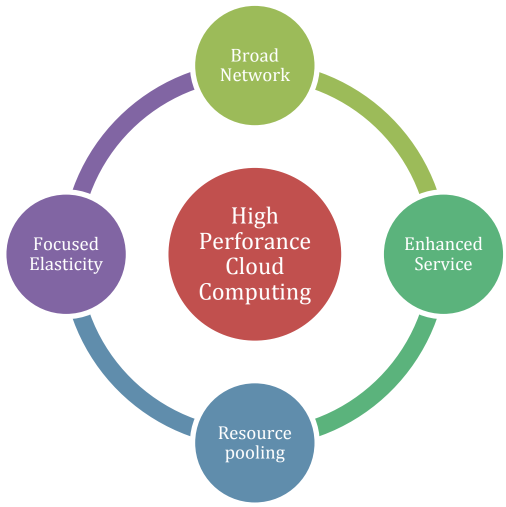

Figure 2. Conceptual model for Cloud Futurology
(Source: [6])

According to Zhou, cloud computing is vehemently promoted by the business giants such as google, amazon and other organisations. The data from Statista effectively demonstrates that the worth of the AI market would cross ninety billion by 2026. It is not a secret cloud computing acts as a tool to maximise the scope and influence in the broader scenario. For example, Google is a brand that tends to utilise cloud systems effectively in the IT industry to maximise its innovation. AI is effectively incorporated into the grain of IT. It assists to manage tasks. Effective employment of AI in the cloud system assists google to maximise its analytical attributes. It also assists in administering the routine processes by the system. It enables the team associated with google to monitor the potency of the cloud. AI also enables google to raise the quality of data management with the effective help of AI [6]. AI tools such as the Google cloud platform assists the team to comprehend the way information is updated and administered. It enables Google to provide more accurate real-life information. It also minimises the risk of fraudulent pursuit. It also can influence vehemently in certain spheres such as consumer service, information regarding supply chain and many others. GCP put forward Iaas, PaaS and SaaS to the different organisations operating in different domains. GCP is basically a service for constructing and manoeuvring applications that can be issued from is HDC. Google cloud made around 13 billion U.S. dollars and it is a testament to the potency of the organisation to employ AI effectively in the cloud system. As the revenue of the google cloud is expanding unfathomably, it can be said that many an organisation is adopting this system at a significant rate [7].

AI as a service is evolving vehemently the way organisation is employing tools and technology. Organisations such as amazon utilise the cloud-based retail website. It enables different brands to vend their product effectively. It also employs pricing attributes that tend to reshuffle the pricing on a given product as per the market demand and the activity of competitor brands. The AI-powered module enables the organisation to efficiently to optimise the price of the product efficiently. Effective employment AI enables the organisation to make proper market analysis and put into the initiative to maintain the sustainability of the organisation.

AI and cloud computing are revolutionising the initiative at every level. Cloud computing is basically an efficient breakthrough tool in the domain of IT. It provides a long range of advantages for the organisation and as well as for the consumers. Despite the advantages, it has some potential challenges such as data breaches. Multiple regression examination and analysis plays a key role to predict the influence of employing cloud computing for different organisations.

III. RESEARCH METHODOLOGY

This study was totally evaluated through a quantitative research approach. The multi regression analysis has been done by using the SPSS tool. According to this statistical method, three different variables are chosen. This topic is based on the influence of AI in cloud computing as well as the effectiveness of AI enabling the wide-range performance of cloud computing. Due to this reason, two independent variables have been chosen. The first one is a year and the second type is the percentage of usage of AI technology in various companies for increasing the performance of cloud computing. An independent variable has also been taken and that is the respective profit percentage of those companies. The usage of AI actually increases the profitability of various companies and the evidence of profitability has been thoroughly discussed in the discussion part.

In large and medium-sized data centers, not only the power of computer hardware such as servers is used, but also the power of additional equipment such as air conditioners. In general, the power consumption of the calculator is approximately the same as the power consumption of other support devices. In other words, if the power consumption of computer hardware in the data is 1, then the power consumption of the entire data center is 2. In some cases the data is very good with a certain durability. technology, electricity consumption can reach a high level. 1.7. , but Google pushed other files up to 1.2 industry leading for well-designed templates. A special feature of this design is the temperature of the data center, which means that the material is included in the operating temperature of the data center.

Virtualization technology is the most important technology for cloud computing, authentication and integration of physical resources. The use of this technology can improve the use of physical therapy equipment, depending on user needs, flexible and fast configuration and transportation. Virtualization technology is completely different from multitasking and hyperthreading technology. Computers are an important part of people's lives. With computer development, important data is created and waiting to be completed; We would have benefited more if this had been done properly, otherwise we would have seen the data broken, the data was just a waste. This study enhances other research activities and promotes conclusions and implementation using the PRISMA approach. The development of computer science will support the development of the organization. Consideration is given to choosing the right and appropriate equipment, including our business. Existing data performance is the largest computer, grid computer, group computer and cloud computing.

The good old ways are always bad for the business environment which has been replaced by the latest technology. The strategy to improve the path needs to be explored in order to have a global impact on the world of intellect, innovation and intellectual property. E-Government in Dubai thrives on the best information in the industry, government support for innovation, financial adequacy and e-government participation of people and businesses. Fake Spy is said to play an important role in the mobile business because it creates user profiles and meets their needs based on their needs and preferences. In addition, the AI team works by identifying and predicting shortcomings, providing a quick solution. Also, AI can communicate with users, for example TOBi is a chatbot developed by Vodafone to help customers answer online questions and support solutions, which makes users' stuff supported.

IV. DISCUSSION AS WELL AS FINDINGS

The first research question is based on the process of enhancing the profitability of various organisations through the incorporation of AI in cloud computing systems. Through using AI various IT organisations reshape their structure totally. In recent days a huge level of competition exists between several IT industries and to make a better competitive advantage those industries implement AI-based computing platforms and incorporate those platforms in their cloud systems to better optimize their resources [9]. Access to unlimited data as well as making effective decisions from those datasets can be effectively done by the AI process. A business organisation required some well-trained analysts whereas AI effectively manage this analysis and can also give better analysis result. Thus, incorporation of AI in cloud computing increases the advantage of those industries in the field of the competitive market. Various statistical data reflects 81% of leaders are planning to invest huge capital for the incorporation of AI. The adoption of AI in several industries enhances automation of digital marketing, CRM and at the same time effective analysis of big data [10].

The second research question is based on the process of enhancement of data security through using AI song with cloud computing. Business organisations incorporate various works in their personalised cloud. Due to this reason, the information must be kept safe always. The management of IT and other business organisation can use AI-powered network security technologies for tracking network traffic [11]. Thus, those organisations can effectively prevent damage to that critical information. Amazon GuardDuty can be considered an important threat detection technology. Maintenance of data privacy is also another important area that can be immensely managed through using artificial intelligence techniques along with cloud systems.

After that, the third research question sheds light on the processes through which AI can enhance the cloud computing system. There are three types of cloud development services and three various kinds of cloud deployment processes. Private cloud, public cloud and hybrid cloud are three different types of cloud processes. Through the help of AI infrastructure, an organisation can create models of machine learning where a broad range of data is applied to definite algorithms and thugs it enables the performance of cloud services by improving data accuracy [12]. In addition to machine languages, the cognitive computing technique also allows users to deliver their personal information. This model can effectively eliminate the correct training model as well as the usage of the proper algorithm model. A lot of benefits can be seen in the case of the usage of AI that also enhances the performance of cloud systems in various organisations. Some of those benefits are mentioned in the following. The development of cloud applications through using AI diminishes the requirement of various hardware and software purchases. Besides that, it can also eliminate the requirement of onsite data centres. Thus, cost-effectiveness is influenced through AI. As cloud computing is based on the total internet of things AI boost up this technology and overall productivity has been increased [13].

In the data analysis part, a statistical analysis has occurred where multi regression analysis has been done. In this analysis, a histogram chart, as well as a scatterplot, has also been figured out. In that area, three things have been chosen for variables and those are profitability, percentage of AI usage in industries and the last one is years. Through this analysis, a significant value reflects .001 that defines the regression model that has been figured out as a good fit for the information that is used to create this analysis.

V. CONCLUSION

After evaluation of the whole paper, it can be concluded that there is a huge need for the implementation of artificial intelligence in the cloud computing processes as it can create a significant level of benefits in various organisations. To create a wide range of competitive advantages it is important to implement AI in various business organisations to smoothen their production processes. Apart from that, to protect big data as well as analysis of those data AI and cloud computing both processes have a huge significance. AI effectively manages and analyses big data whereas the cloud process maximises the security of those data. There are various machine learning tools and cognitive computing techniques through which one can easily increase the performance of big data. In addition to this cost-effectiveness can also be increased as there is less need for maintenance costs and online data centres. At last, it can be stated that AI has a huge influence on increasing the performance of cloud computing that also assist in the enhancement of profitability of an organisation.

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